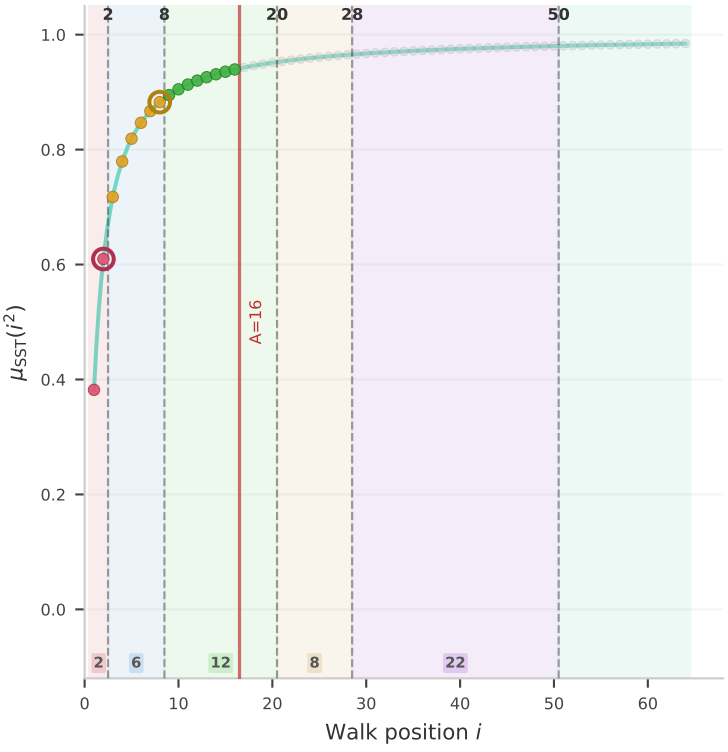
Shell Structure from the Sierpiński Walk

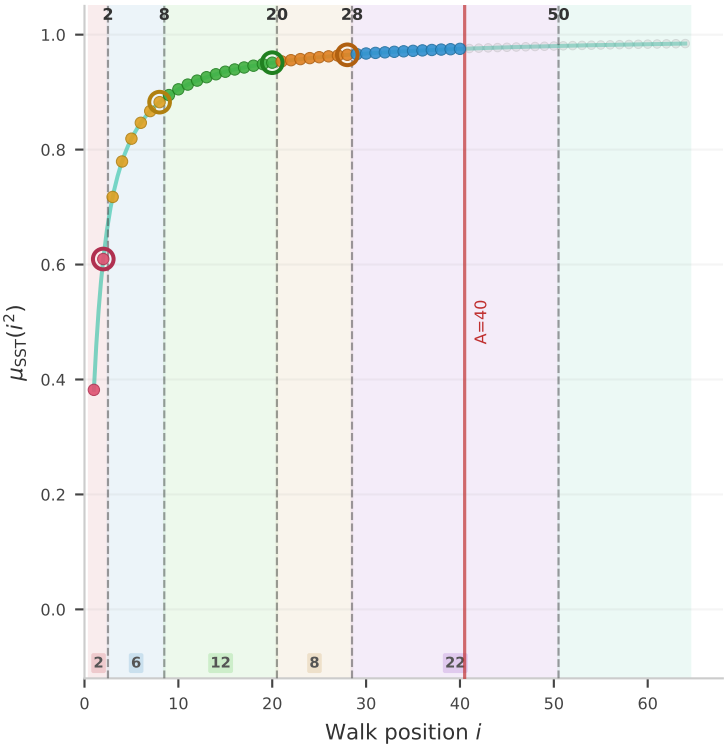
$M(n) = 2\,[T(n) - G(n)]$ $T(n) = n(n+1)(n+2)/6$ $G(n) = (n-1)n/2$ $(n \geq K)$

O-16 (Z=8)



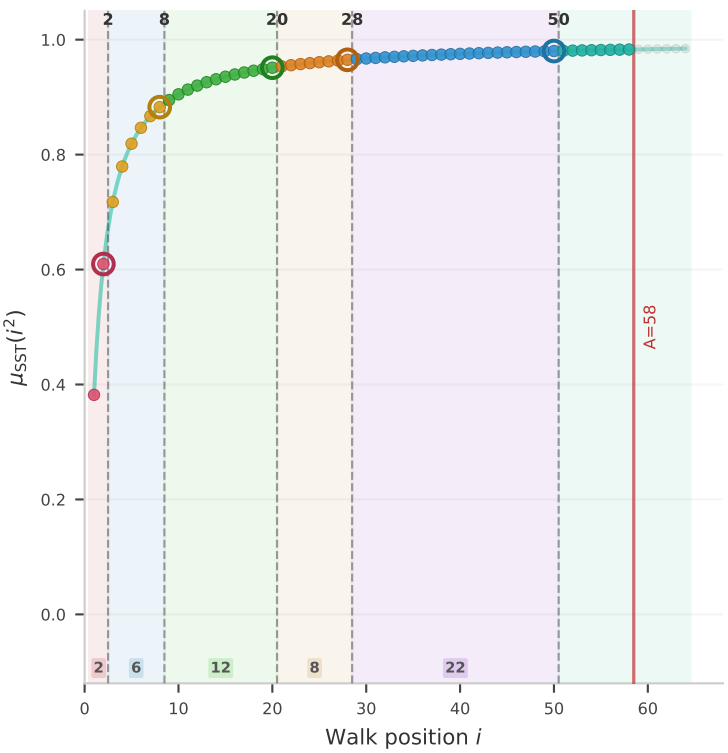
B/A = 7.976 Predicted = 7.755 Error = 2.77%

Ca-40 (Z=20)



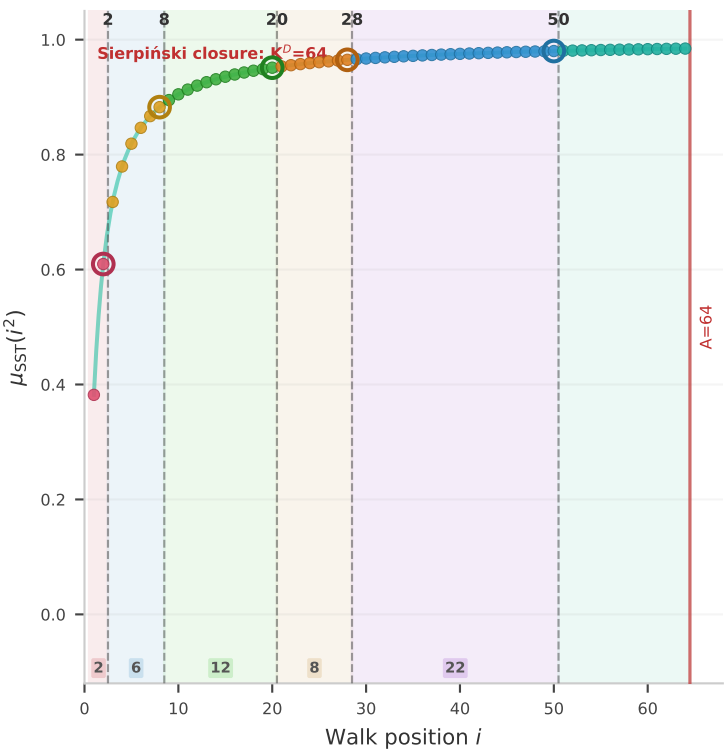
B/A = 8.551 Predicted = 8.507 Error = 0.52%

Ni-58 (Z=28)



B/A = 8.732 Predicted = 8.710 Error = 0.25%

Zn-64 (Z=30)



B/A = 8.756 Predicted = 8.756 Error = 0.00%

Shell capacities: 2 + 6 + 12 + 8 + 22 + 32 + 44

Magic numbers: 2, 8, 20, 28, 50, 82, 126

The 20→28 gap emerges from gate exclusion at $n = K = 4$, without spin-orbit coupling